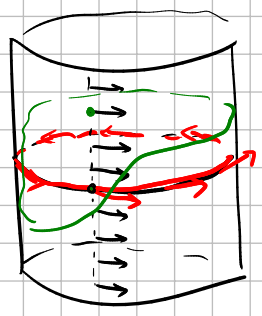


$\pi^{-1}(U) \to U \times F$

looking at the rep of $U(1)$ on $\mathbb{C}^2$ given by

$\rho : z \in U(1) \mapsto \rho(z) \in \mathrm{Aut}(\mathbb{C}^2)$

$\rho(z)\, v = z v$ $\qquad$ $\rho(z_1 z_2) = \rho(z_1) \circ \rho(z_2)$

$\rho(z)^{-1} = \rho(z^{-1})$

$P \times_{U(1)} \mathbb{C}^2 = P \times \mathbb{C}^2 / U(1)$

action of $U(1)$ on $P \times \mathbb{C}^2$: $\quad (u, v) \triangleleft g = (u \triangleleft g, \rho(g^{-1}) v)$

$= (u \triangleleft g, g^{-1} v)$

$u \in P$
$v \in \mathbb{C}^2$
$g \in U(1)$

$((u, v) \triangleleft g) \triangleleft h = (u \triangleleft gh, (gh)^{-1} v)$

aside: G, H groups

G acts on H by $h \triangleleft g = hg$

right coset Hg (orbit by action of g)

$E = P \times \mathbb{C}^2 / U(1) = \{ [(u, v)] \mid u \in P, v \in \mathbb{C}^2 \}$

$[(u, v)] = \{ (u, v) \triangleleft g \mid g \in U(1) \}$ <u>orbits</u>

$\pi_E : E \to M \qquad \pi_E([(u, v)]) = \pi(u)$

$\psi_i : M \to E$ satisfy $\pi_E \circ \psi_i = \mathrm{id}_M$

$\psi_i(x) = [(u, v)]$ must have $\pi(u) = x$

way to look at something related to ψ_i that associates a vector in $\mathbb{C}^2$ to each point in M

---

options for understanding $\psi_1 \wedge \psi_2$

① $\psi_1 \wedge \psi_2 = \psi_1 \otimes \psi_2 - \psi_2 \otimes \psi_1$

$\mathrm{rank}_{\mathbb{C}}(E) = 2$

if a vector bundle of rank n has n global nonzero-zero, independent sections then it's trivial

$\rightarrow \quad P \times_{U(1)} \mathbb{C}^2 \cong M \times \mathbb{C}^2$ (it's possible ψ_1 and ψ_2 are not independent)

---

assume $E = M \times \mathbb{C}^2$

---

if V n-dimensional vector space $\quad \Lambda^k V = \mathrm{span}\{ s_1 \wedge s_2 \dots \wedge s_k \mid s_1, \dots, s_k \in V \}$

max k is n $\quad \dim(\Lambda^k V) = \binom{n}{k}$

$u \wedge v = u \otimes v - v \otimes u$ bilinear map

$(\alpha u) \wedge v = \alpha (u \wedge v)$

$(u_1 + u_2) \wedge v = u_1 \wedge v + u_2 \wedge v$

$u \wedge v = -v \wedge u$

---

bivectors

$\Lambda^2 V = \mathrm{span}\{ e_i \wedge e_j \mid i,j = 1, \dots, n \}$ $\quad \{e_i\}$ basis of V

if $\dim(V) = 2$ $\quad \Lambda^2 V = \mathrm{span}\{ e_1 \wedge e_2 \}$ $\quad \dim(\Lambda V)$

$\det(E) \cong M \times \Lambda^2 \mathbb{C}^2 \cong M \times \mathbb{C}$

since ψ_1 ψ_2 are such that $\{\psi_1|_x, \psi_2|_x\}$ basis of E_x

$\psi_1 \wedge \psi_2$ is a global nowhere-zero section of $\det(E)$

as a vector bundle

$f : M \to \mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ $\qquad$ $\det(E) \cong M \times \mathbb{C}$

$\psi_1 \wedge \psi_2 |_x \in \Lambda^2 \mathbb{C}^2 \xrightarrow{\Psi} \mathbb{C}$

$f(x) = \Psi(\psi_1 \wedge \psi_2 |_x)$

$f : M \to \mathbb{C}^\times$ smooth

$\arg f = ?$

$f(x) = r(x) e^{i\theta(x)}$ $\qquad$ $\theta \in [-\pi, \pi)$

$\theta(x) = (\arg f)(x)$

$d\theta$ might make sense

$M = \mathbb{R}$ $\qquad$ $\theta(x) = x$

$$\arg(e^{i\theta(x)}) = \begin{cases} x & \text{if } x \in [-\pi, \pi) \\ x - 2\pi & \text{if } x \in [\pi, 3\pi) \\ x - 4\pi & \text{if } x \in [3\pi, 5\pi) \\ \vdots & \end{cases}$$

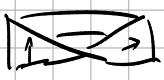
$$\frac{d}{dx} \arg(e^{i\theta(x)}) = 1$$

$U \subseteq M$ simply conn. any parallel section τ of $\det(E)$ is unif. up to constant phase

---

$M = M_t \times M_s \quad \Rightarrow \quad TM \cong TM_t \oplus TM_s$

$\not\Leftarrow$



E vector bundle

global section of E: $s : M \to E$

that to each $x \in M$ associates a vector in E_x

---

a local frame for E is a set of $\dim(E)$ local fields $s_i : U \subseteq M \to E$ such that at each point $x \in U$, $\{s_i(x)\}$ basis of E_x

---

if E has a global frame then E is parallelizable (trivial)

$E \cong M \times F$

$\exists$ map $\Psi : E \to M \times F$ diffeomorphism and preserves fibre

$\mathrm{proj}_1 \circ \Psi = \pi$

$$\begin{tikzcd} E \arrow[r, "\Psi"] \arrow[d, "\pi"'] & M \times F \arrow[dl, "\mathrm{proj}_1"] \\ M & \end{tikzcd}$$

map $v \in E_x \mapsto \Psi(v) \in \{x\} \times F$ linear

- assume θ is a clock (global nowhere-vanishing 1-form)

induces a split $TM = \mathrm{span}(T) \oplus S$ $\qquad S = \ker \theta$

$\rightarrow$ T s.t. $\theta(T) > 0$

$T : M \to TM$, $x \mapsto T_x$ $\qquad \theta_x(T_x) := 1$

use duality between T^*M and TM to get vector field out of θ

---

assume $[\Psi] : M \to \mathbb{CP}^1$ smooth $\qquad$ what is $[\Psi]$?

$\Omega = [\Psi]^* \omega_{FS}$ $\qquad \sigma = \Omega|_S \in \Omega^2(S)$

need Ω to be non-vanishing

(need $[\Psi]$ injective)

$V = \Omega \wedge d(\iota_T \Omega)$ $\qquad$ 4-form

A.1 in local coords t, x, y, z on $U \subseteq M$ s.t. $\theta = dt$ $\quad T = \partial_t$

$$\Omega = A\, dt \wedge dx + B\, dt \wedge dy + C\, dt \wedge dz + D\, dx \wedge dy + E\, dx \wedge dz + F\, dy \wedge dz$$

$\boxed{\text{assume } \Omega \wedge \Omega = 0}$ $\rightarrow$ $dt \wedge dx \wedge dy \wedge dz \, (\underbrace{AF - BE + CD}_{=0})$

$$\iota_T \Omega = A\, dx + B\, dy + C\, dz$$

$$d(\iota_T \Omega) = (\partial_t A)\, dt \wedge dx + (\partial_t B)\, dt \wedge dy + (\partial_t C)\, dt \wedge dz$$

$$\underbrace{(\partial_x B - \partial_y A)}_{U}\, dx \wedge dy + \underbrace{(\partial_x C - \partial_z A)}_{V}\, dx \wedge dz + \overbrace{(\partial_y C - \partial_z B)}^{W}\, dy \wedge dz$$

$$\Omega \wedge d(\iota_T \Omega) = dt \wedge dx \wedge dy \wedge dz \left(AW - BV + CU + D(\partial_t C) - E(\partial_t B) + F(\partial_t A) \right)$$

is this zero anywhere?

maybe must be $\begin{cases} AW - BV + CU = 0 \\ D(\partial_t C) - E(\partial_t B) + F(\partial_t A) = 0 \end{cases}$

for a generic immersion $[\Psi]$ these requirements are never true on any open set not sure it's true, assume it is

---

independence from choice of T

if $T' = T + X$ with $\theta(X) = 0$

$$\Omega \wedge d(\iota_{T'}\Omega) = \Omega \wedge d(\iota_T \Omega + \iota_X \Omega)$$

$d(\iota_X \Omega)$ is a 2-form taking values in $\ker \Omega$ ← why?

$\iota_X \Omega$ spatial 1-form (?) $\Omega \wedge \iota_X \Omega = 0$ because $\Omega \wedge \Omega = 0$

$$0 = d(\Omega \wedge \iota_X \Omega) = d\Omega \wedge \iota_X \Omega + \Omega \wedge d(\iota_X \Omega)$$

$[\Psi]$ must be related to θ

---

need $d[\Psi](T) = 0$ NO $d[\Psi]$ is injective (kern

$d[\Psi]_a(T_a) = 0$

$$\Omega = [\Psi]^* \omega_{FS}$$

$$(\iota_T \Omega)_a (v) = \omega_{FS}|_{[\Psi](a)} \big(\underbrace{d[\Psi]_a(T_a)}_{=0}, d[\Psi]_a(v) \big) = 0$$

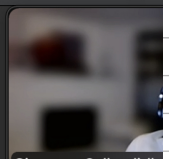
---

need $\Omega \wedge d(\iota_X \Omega) = 0$ if $\theta(X) = 0$

$\cancel{\iota_T \Omega = \theta}$? maybe? $\iota_T \Omega(T) = 0$

$\iota_X \Omega = 0$? $\theta(T) = 1$

4. Explicit, rigorous definition of $[\Psi]$ tied to Paper 2



In Paper 2 you already have:

- A rank-2 Hermitian bundle $E \to M$.
- A "pure state" section $\Psi \in \Gamma(E)$ with $\|\Psi(x)\| = 1$ in each fiber.
- The density operator field $\rho(x) = |\Psi(x)\rangle\langle\Psi(x)|$.

At each point $x \in M$, the fiber E_x is a 2D complex Hilbert space, isomorphic to $\mathbb{C}^2$.

The set of **pure states** in that fiber is the projective space:
$\mathbb{P}(E_x) \cong \mathbb{C}\mathbb{P}^1$.

So you naturally define:

$$[\Psi] : M \to \mathbb{C}\mathbb{P}^1, \quad x \mapsto \text{ray of } \Psi(x),$$

i.e. the equivalence class of $\Psi(x)$ modulo nonzero complex scalar.

In a local orthonormal frame (e_1, e_2) of E:
$\Psi(x) = z_1(x)e_1(x) + z_2(x)e_2(x)$,

then
$[\Psi](x) = [z_1(x) : z_2(x)] \in \mathbb{C}\mathbb{P}^1$.

This is **exactly** the usual identification of a normalized 2-component spinor with a point on the Bloch sphere.

From there, the Fubini–Study 2-form ω_{FS} on $\mathbb{C}\mathbb{P}^1$ is the **unique** U(2)-invariant Kähler form, and physically it's the "distinguishability metric" on pure states (same thing that appears in quantum Fisher information, etc.).